

SHUBHAM SAHU

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PROFESSIONAL SUMMARY

Electronics & Computer Engineering student with 12 shipped open-source projects across agentic AI, applied ML, and secure full-stack products — including a multi-agent AML platform (199/199 backend tests passing), a fully local AI recruitment pipeline (31 REST endpoints), and an encrypted document-sharing platform live on web and the Play Store.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, Dart, Rust
- **AI / ML:** LangGraph, agentic AI workflows, XGBoost, SHAP (explainable ML), YOLOv8, OpenCV, Whisper, scikit-learn, Ollama (local LLMs), embeddings & semantic search, prompt engineering
- **Backend & Data:** FastAPI, Node.js/Express, Flask, PostgreSQL (Supabase: RLS, triggers, Edge Functions), Firebase, Redis/BullMQ, SQLite, REST API design
- **Frontend:** React, Next.js, Flutter, Vite, Tailwind CSS
- **DevOps & Security:** Docker, GitHub Actions CI/CD, Vercel, Render, AES-256 client-side encryption, RBAC, Row-Level Security, homomorphic encryption (TFHE-rs)

EXPERIENCE

Autonomous Robotics Intern — MakerWorks Lab, Mumbai Jun 2026 – Jul 2026

- Built R3Bin, an AI waste-sorting robot: designed the full ROS 2 node graph (camera detection → touchscreen HMI → servo control → ESP32 bridge) and integrated a YOLOv8 vision model classifying waste in real time
- Designed a human-in-the-loop correction flow logging every AI decision and override to Supabase, plus a hardware-free simulation harness so the full pipeline can be validated without a camera or physical waste

Embedded Coding Intern — Fostride, Mumbai Dec 2025 – Feb 2026

- Developed and tested ESP32 firmware for IoT and automation hardware; reusable driver modules reduced integration time ~20%
- Implemented I2C/UART sensor integration enabling real-time data pipelines across 3+ hardware configurations

AI & Agile Intern — Practice Agile Consultancy, Mumbai Jun 2025 – Jul 2025

- Co-developed agentic AI solutions and contributed to an AI Fundamentals course that onboarded 200+ learners; completed 2 Scrum sprint cycles on live client projects and authored 3 published technical assets

Head of Product Development — Robotics & Automation Club, TSEC Mumbai Feb 2025 – Jul 2026

- Led a 15-member cross-functional team through 3+ product builds from ideation to working prototype, delivering every club showcase on time

PROJECTS

crypt.ml — Multi-Agent AML/CFT Compliance Platform (Python, FastAPI, XGBoost, SHAP, NetworkX, React, Docker)

- Designed an event-driven 5-agent architecture (Rules, NLP, ML, Graph, Orchestrator) coordinating over an async EventBus to reach a weighted risk consensus on every transaction
- Shipped explainable risk scoring (XGBoost + per-score SHAP explanations), graph-based laundering detection (Louvain communities, cycle traversal), and automated Suspicious Activity Report drafting; 199/199 backend tests, one-click Docker deploy

ResumeDB — Local-First AI Recruitment Platform (Python, FastAPI, React, Ollama, Whisper, OpenCV)

- Built the full hiring pipeline — CV parsing with OCR → embedding-based semantic search → auto-graded AI assessments → live AI interview with real-time behavioral analysis — exposed via 31 REST endpoints
- Ran 100% of LLM inference locally via Ollama, eliminating cloud API cost and candidate-data privacy risk; scored interviews with Whisper transcription and OpenCV posture/gaze/emotion analysis

NON_SUS — Encrypted Document-Sharing Platform (Flutter, Supabase, Rust/TFHE-rs) — live: nosus.foo + Play Store

- Architected a 4-part monorepo shipping web and Android from one codebase; built SecureSend links with expiry, view limits, dynamic recipient watermarking, and real-time audit logs
- Implemented zero-knowledge burn notes (client-side AES-256, key carried only in the URL fragment) and a feature-flagged homomorphic-encryption microservice (Rust, TFHE-rs); Row-Level Security on every table, CI/CD for signed Android builds

Honeypot Agent — Agentic AI Scam-Defense System (Python, FastAPI, Groq)

- Built an LLM scam-intent classifier gating a persona-driven agent that engages scammers indefinitely while extracting structured intelligence (contacts, financial details, links); deployed on Render with a full reasoning-trace admin dashboard

EDUCATION

B.E. Electronics & Computer Engineering — Thakur Shyamnarayan Engineering College (TSEC), Mumbai Expected 2028

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900), Jan 2026 • Career Essentials in Generative AI — Microsoft & LinkedIn, Jan 2026 • Applied Python for Data Science — TSEC, Jan 2026